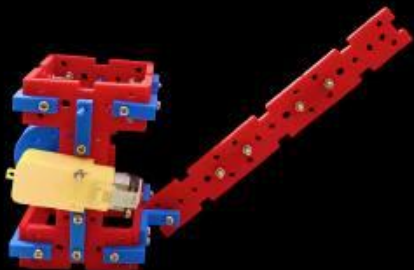


level 2

no. of sessions : 4

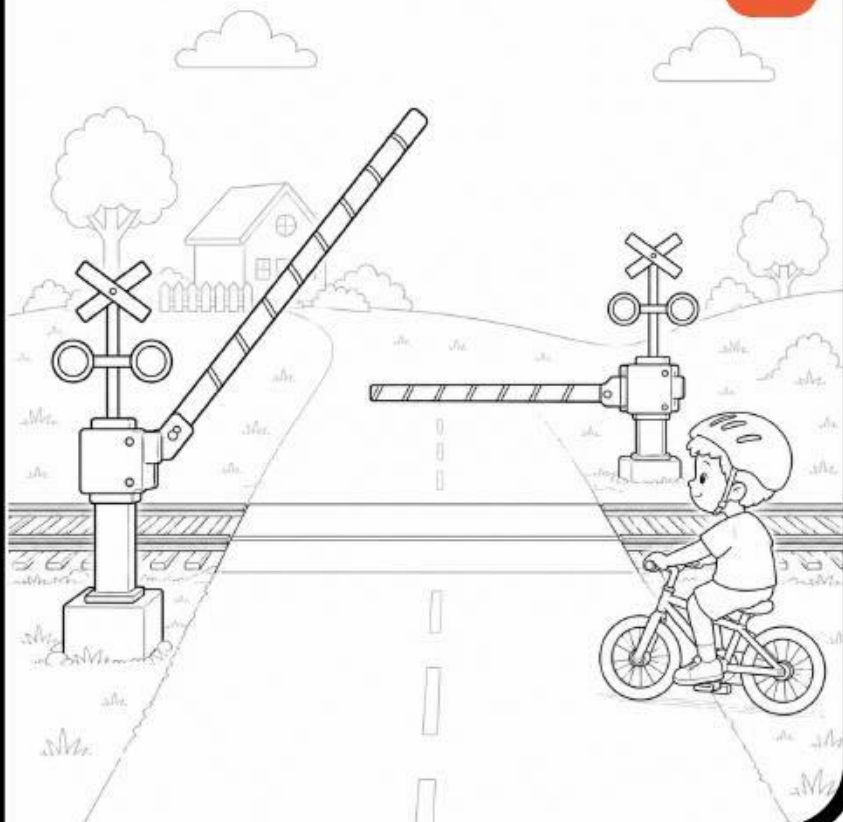
railway barrier

The railway barrier automatically opens and closes to stop vehicles when a train is coming, just like the barriers at railway crossings.



simple machine: levers

electronic parts: open-close circuit





Motor Driver

The **brain of a robot**. I listen to input devices like sensors and tell output devices like motors/LEDs when to turn ON/OFF.



IR Sensor

I am like a **robot's eyes**. I help robots see objects or follow a path without touching anything.



LDR Sensor

I am like a **light detector** who tells a robot if it is bright or dark and i am used in automatic lights.



Motor

I **change electricity into movement** and help wheels, fans, and robots spin and move.



DPDT Switch Controller

Like a **direction button** - i make a motor spin forward or backward without using coding.



Servo Motor

I am a **smart motor** that can stop at the exact position you want. I am used in robot arms, gates, and moving parts.



Servo Controller

I am like a **control room** for the servo motor and lets you move the servo easily to test how it works.



Battery & Battery Holder

I **give power** to the project, and my battery holder keeps the battery safe and connected. Together, we help the model work.



Jumper Wires

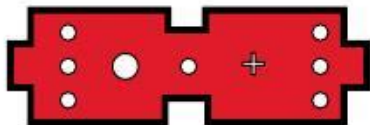
We are the **roads for electricity** who connect the different parts of the circuit, so power can travel.



Motor Clamp

I am like a **strong hand** that holds the motor tightly in place while it spins.

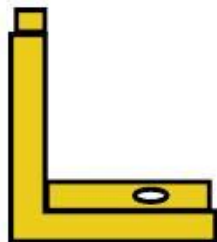
materials needed



M-M beam - x4



F-F beam - x4



L clamp - x10



small screw - x18



nut - x18

step 1 : assemble the base



1. square 1

Join 2 x F-F Beams and 2 x M-M Beams together to form a square.

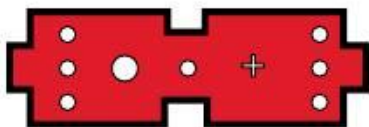
Use 4 x L Clamps to hold them together.



2. square 2

Add 2 L clamps to square as shown.

materials needed



M-M beam - x2



small screw - x6

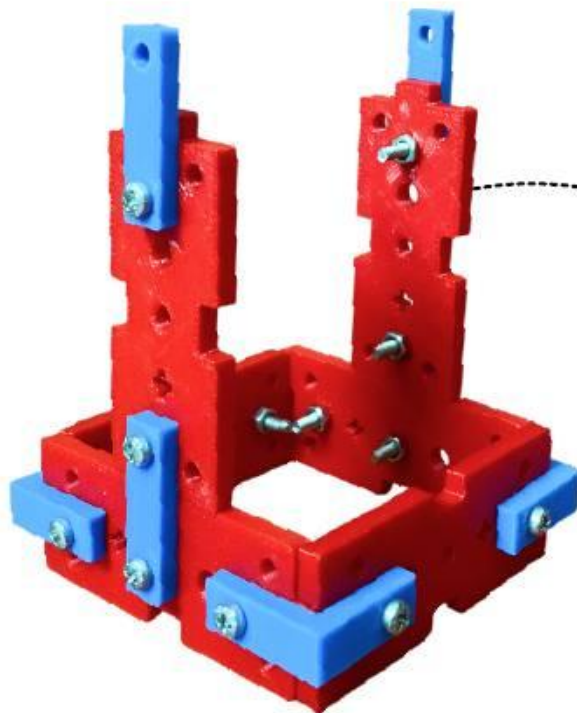


nut - x6



I clamp - x4

step 2: assemble the supports



Attach 2 M-M beams
1 clamps to square 1

materials needed



motor - x1



long screw - x1



nut - x1



coupler - x1

step 3: attach the motor and motor coupler



1.

Fix the **motor** to the side of the support using a long screws



2.

Fix the **motor coupler** to the inner axle of the motor

materials needed



M-F beam - x2



F-F beam - x1



I clamp - x2

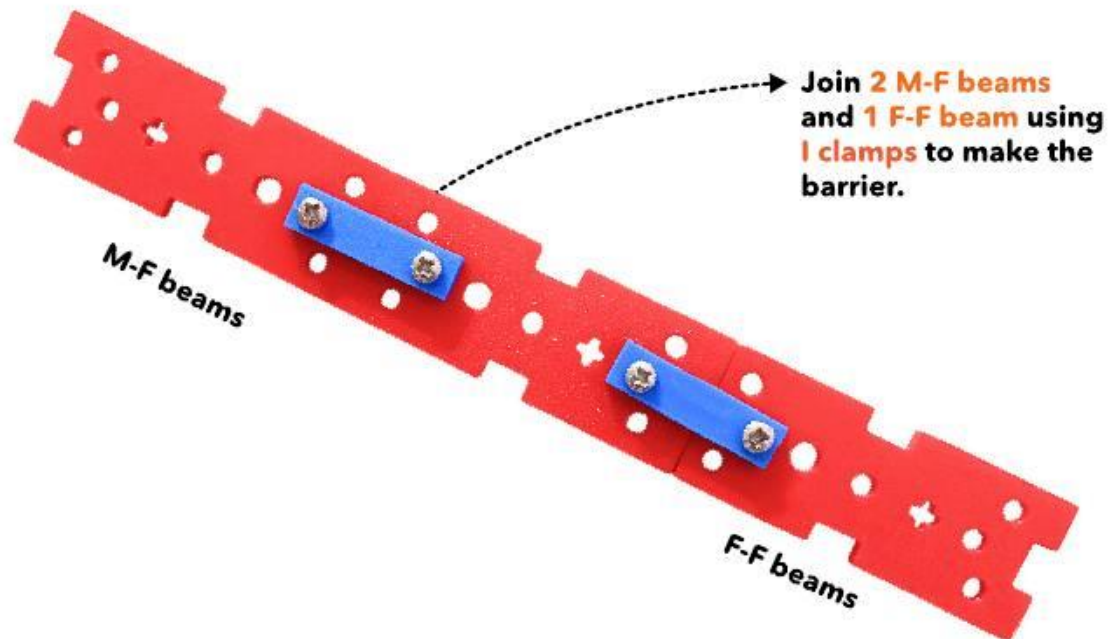


small screw - x4



nut - x4

step 4: assemble the barrier



materials needed

long axle x1



rope - 10cm - x1



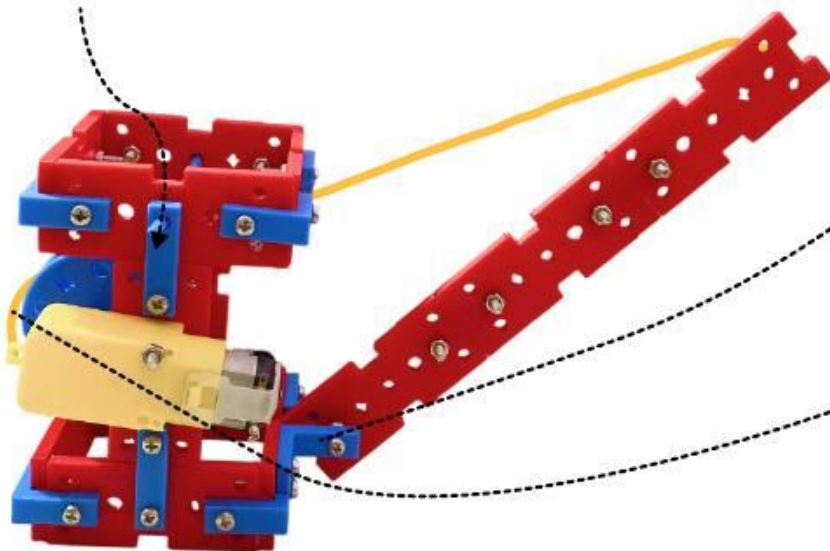
long screw - x1



nut - x1

step 5: attach the barrier

1. Pass the axle through the **second square** on top



2.

Fix the barrier in between the L clamps of the first square using long screws.

3.

Fix a rope to the motor coupler and the other end to the beam of the barrier.

materials needed



motor controller - x1



battery holder x1



battery x1

step 7: circuit

